

Faculty of Computing

Year 2 Semester 1 (2026)

IT2011 - Artificial Intelligence and Machine Learning

Progress Review - I

You are required to attend the Progress Review I Viva, which focuses on your Data Cleaning, Preprocessing, and Exploratory Data Analysis (EDA), applied to your assigned dataset.

Progress Review – Viva Details

- Mode: In-class (during your regular lab session)
- Duration: Approximately 10 minutes per student
- Format: Viva-style Q&A + brief walkthrough of your notebook/script

Individual Requirement (25 marks per student):

- Each student must handle and present following preprocessing techniques:
 - Handling missing data
 - Encoding categorical variables
 - Outlier removal
 - Normalization / scaling
 - Feature engineering (Feature selection and dimension reduction)
- Each student must
 - Clearly explain each technique
 - Justify why it was needed for their dataset
 - Show the implementation (code/output) for each technique
 - Present at least three EDA visualizations (e.g., histogram, boxplot, correlation matrix) and interpret each one

Students who do not present or fail to demonstrate complete individual implementation will receive zero.

Repository Layout for Group Deliverable folder (Upload a Zip file to Courseweb)

Student_ID/

- README.md — Overview of the project, assigned dataset details, group member roles, how to run code
- data/raw/ — Assigned dataset(s) as provided
- data/external/ — Any external reference datasets (if used)
- notebook.ipynb — named with IT Number & technique
- results/eda_visualizations/ — Plots & charts (PNG/JPEG)
- results/logs/ — Any logs from execution (optional)
- results/outputs/ — Final processed dataset / features